

TEACHER: BOHL CVEK FINS KERJ MCLS PHIN ROSP SCHP

STUDENT NAME: _____

YEAR 12 MATHEMATICAL METHODS

Test (Tech-Free)

Test theme: Differential Calculus

Writing time: 55 minutes

QUESTION AND ANSWER BOOK

Structure of book

<i>Number of questions</i>	<i>Number of questions to be answered</i>	<i>Number of marks</i>
5	5	36

- Students are permitted to bring into the test assessment area: pens, pencils, highlighters, erasers, sharpeners and rulers.
- Students are NOT permitted to bring into the test assessment area: any technology (calculators or software), notes of any kind, blank sheets of paper and/or correction fluid/tape.

Materials supplied

- Question and answer book of 8 pages.
- Working space is provided throughout the book.

Instructions

- Write your **name** in the space provided above on this page.
- Unless otherwise indicated, the diagrams in this book are **not** drawn to scale.
- All written responses must be in English.

Students are NOT permitted to bring mobile phones and/or any other unauthorised electronic devices into the assessment area.

Instructions

Answer **all** questions in the spaces provided.

Unless otherwise specified, an **exact** answer is required to a question.

In questions where more than one mark is available, appropriate working **must** be shown.

Question 1 (8 marks)

Let the function $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^3 - 5x$.

$A(-1, f(-1))$ and $B(1, f(1))$ are two points on the graph of $f(x)$.

- a. i.** Show that the coordinates of A and B are $(-1, 4)$ and $(1, -4)$ respectively. 1 mark

- ii.** Find the distance AB . 1 mark

- b.** Find the equation of the straight line that passes through points A and B . 2 marks

Let the function $g: R \rightarrow R, g(x) = x^3 - px, p \in R^+$.

Let $C(-1, g(-1))$ and $D(1, g(1))$ be two points on the graph of $g(x)$.

- c. Find the distance between C and D in terms of p . 2 marks

- d. Find the value(s) of p such that the distance between C and D is equal to $p + 1$. 2 marks

Question 2 (2 marks)

Find the derivative of the following:

- a. $y = (3x^2 - 5x)^5$. 1 mark

- b. $g(x) = \frac{1}{2x-4} + 3$. 1 mark

Question 3 (7 marks)

Find the derivative of the following:

a. $y = \frac{e^{3x} + e}{x^2 - 3}.$ 2 marks

b. $y = e^{2x} \sin\left(\frac{x}{2} + \pi^2\right).$ 2 marks

c. $y = \sqrt{x} \log_e(\cos(2x)).$ 3 marks

Question 4 (4 marks)

- a. Let $f(x) = \frac{1}{e^{x^2}}$. Find $f'(-3)$. 2 marks

- b. Let $g(x) = \log_e(\tan(2x))$. Find $g'\left(\frac{\pi}{8}\right)$. 2 marks

Question 5 (3 marks)

Find the derivative of $(1 - x^2)\sqrt{3x - 2}$ in the form $\frac{ax^2 + bx + c}{2\sqrt{3x - 2}}$ where a , b and c are integers.

Question 6 (12 marks)

Let the function $h:[0,\infty)\rightarrow R$, $h(t)=(t-1)^2e^{-t}$.

- a.

The rule for the derivative of h can be expressed in the form $h'(t)=(-t^2+bt+c)e^{-t}$ where b and c are integers. Show that $b=4$ and $c=-3$.

2 marks

- b.

Find the equation of the tangent to $y=h(t)$ at the point where $t=2$.

2 marks

- c. The graph of $y = h(t)$ has stationary points $(1, p)$ and (m, n) where p, m and n are integers. Find the exact values of p, m , and n .

2 marks

- d. Find the equation of the normal to $y = h(t)$ at the point where $t = 1$.

1 mark

- e. Let the function $s : [0, \infty) \rightarrow \mathbb{R}$, $s(t) = 2h(t) - 5$.

State the coordinates of the stationary points of s .

2 marks

- Find the value(s) of k such that the graph of $y = f(t)$ has exactly one stationary point. 3 marks

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TURN OVER

BONUS QUESTION (3 marks)

Find the value of a and b , where $0 < b < \frac{\pi}{2}$, such that the normal to the curve $y = a \sin(bx)$ at the point where $x = 4$ is $y = \frac{6}{\pi}x - \frac{24}{\pi}$.
